



FUNDAMENTALS OF COMMUNICATIONS

COURSE PROJECT REPORT

DR. NIDA

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Introduction

In this report, we present the steps taken to design and implement our network topology using Cisco Packet Tracer. The objective was to create a fully functional enterprise network with proper IP addressing, routing, DHCP configuration, VLANs, and secure wireless connectivity. Our design is based on two main routers (Router 1 and Router 2) interconnected via serial links.

Topology

Our network topology consists of:

- Two Routers (Router 1 and Router 2) configured with static IP addresses and connected via serial links for WAN communication.
- Multiple Switches: Switch 1 connected to Router 1 and Switch 2 connected to Router 2.
- PCs and Printers: Connected to the respective switches for each router.
- DHCP Server: Configured to provide dynamic IP addressing for certain devices.
- Access Point: Providing secure wireless connectivity for mobile devices.
- IP Phones: Configured and connected via switches for VoIP functionality.

[Video simulation](#)

[PacketTracer File](#)

Table of addresses

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router 1	Ge 0/0	10.10.10.30	255.255.255.252	N/A
	Ge 0/1	10.10.10.45	255.255.255.252	N/A
	Ge 0/2	10.10.10.53	255.255.255.252	N/A
	Se 0/2/0	10.10.10.26	255.255.255.252	N/A
	Se 0/2/1	10.10.10.38	255.255.255.252	N/A
Router 2	Ge 0/0	10.10.10.22	255.255.255.252	N/A
	Ge 0/1	10.10.10.41	255.255.255.252	N/A
	Ge 0/2	10.10.10.49	255.255.255.252	N/A
	Se 0/2/0	10.10.10.34	255.255.255.252	N/A



	Se 0/2/1	10.10.10.37	255.255.255.252	N/A
Switch 0	Ge 1/0/1 – Ge 1/0/24 Ge 1/1 – Ge 1/4	N/A	N/A	N/A
Switch 1	Ge 1/0/1 – Ge 1/0/24 Ge 1/1/1 – Ge 1/1/4	N/A	N/A	N/A
Switch 2	Ge 1/0/1 – Ge 1/0/24 Ge 1/1/1 – Ge 1/1/2	N/A	N/A	N/A
Switch 3	Fa 0/1 – Fa 0/24 Ge 0/1 – Ge 0/2	N/A	N/A	N/A
Switch 4	Fa 0/1 – Fa 0/24 Ge 0/1 – Ge 0/2	N/A	N/A	N/A
Switch 5	Fa0/1 - Fa0/24 Ge 0/1 – Ge 0/2	N/A	N/A	N/A
PC0	FastEthernet0	192.168.11.140	255.255.255.192	192.168.11.129
PC1	FastEthernet0	192.168.11.145	255.255.255.192	192.168.11.129
PC2	FastEthernet0	192.168.11.143	255.255.255.192	192.168.11.129
PC3	FastEthernet0	192.168.11.147	255.255.255.192	192.168.11.129
PC4	FastEthernet0	192.168.11.149	255.255.255.192	192.168.11.129
PC5	FastEthernet0	192.168.11.137	255.255.255.192	192.168.11.129
PC6	FastEthernet0	192.168.11.141	255.255.255.192	192.168.11.129
PC7	FastEthernet0	192.168.11.144	255.255.255.192	192.168.11.129
PC8	FastEthernet0	192.168.11.135	255.255.255.192	192.168.11.129
PC9	FastEthernet0	192.168.11.148	255.255.255.192	192.168.11.129
PC10	FastEthernet0	192.168.11.146	255.255.255.192	192.168.11.129
PC11	FastEthernet0	192.168.11.201	255.255.255.192	192.168.11.193
PC12	FastEthernet0	192.168.11.206	255.255.255.192	192.168.11.193
PC13	FastEthernet0	192.168.11.209	255.255.255.192	192.168.11.193
PC14	FastEthernet0	192.168.11.210	255.255.255.192	192.168.11.193
PC15	FastEthernet0	192.168.11.208	255.255.255.192	192.168.11.193
PC16	FastEthernet0	192.168.11.218	255.255.255.192	192.168.11.193
PC17	FastEthernet0	192.168.11.211	255.255.255.192	192.168.11.193
PC18	FastEthernet0	192.168.11.220	255.255.255.192	192.168.11.193
PC19	FastEthernet0	192.168.11.212	255.255.255.192	192.168.11.193
PC20	FastEthernet0	192.168.11.205	255.255.255.192	192.168.11.193
PC21	FastEthernet0	192.168.12.17	255.255.255.192	192.168.12.1
PC22	FastEthernet0	192.168.12.21	255.255.255.192	192.168.12.1
PC23	FastEthernet0	192.168.12.14	255.255.255.192	192.168.12.1
PC24	FastEthernet0	192.168.12.10	255.255.255.192	192.168.12.1
PC25	FastEthernet0	192.168.12.8	255.255.255.192	192.168.12.1
PC26	FastEthernet0	192.168.12.22	255.255.255.192	192.168.12.1
PC27	FastEthernet0	192.168.12.23	255.255.255.192	192.168.12.1
PC28	FastEthernet0	192.168.12.24	255.255.255.192	192.168.12.1
PC29	FastEthernet0	192.168.12.18	255.255.255.192	192.168.12.1
PC30	FastEthernet0	192.168.12.16	255.255.255.192	192.168.12.1
PC31	FastEthernet0	192.168.12.7	255.255.255.192	192.168.12.1



PC32	FastEthernet0	192.168.12.10	255.255.255.192	192.168.12.1
PC33	FastEthernet0	192.168.12.77	255.255.255.192	192.168.12.65
PC34	FastEthernet0	192.168.12.81	255.255.255.192	192.168.12.65
PC35	FastEthernet0	192.168.12.87	255.255.255.192	192.168.12.65
PC36	FastEthernet0	192.168.12.84	255.255.255.192	192.168.12.65
PC37	FastEthernet0	192.168.12.91	255.255.255.192	192.168.12.65
PC38	FastEthernet0	192.168.12.71	255.255.255.192	192.168.12.65
PC39	FastEthernet0	192.168.12.78	255.255.255.192	192.168.12.65
PC40	FastEthernet0	192.168.12.90	255.255.255.192	192.168.12.65
PC41	FastEthernet0	192.168.12.80	255.255.255.192	192.168.12.65
PC42	FastEthernet0	192.168.12.72	255.255.255.192	192.168.12.65
PC43	FastEthernet0	192.168.12.141	255.255.255.192	192.168.12.129
PC44	FastEthernet0	192.168.12.157	255.255.255.192	192.168.12.129
PC45	FastEthernet0	192.168.12.156	255.255.255.192	192.168.12.129
PC46	FastEthernet0	192.168.12.140	255.255.255.192	192.168.12.129
PC47	FastEthernet0	192.168.12.142	255.255.255.192	192.168.12.129
PC48	FastEthernet0	192.168.12.139	255.255.255.192	192.168.12.129
PC49	FastEthernet0	192.168.12.137	255.255.255.192	192.168.12.129
PC50	FastEthernet0	192.168.12.138	255.255.255.192	192.168.12.129
PC51	FastEthernet0	192.168.12.154	255.255.255.192	192.168.12.129
PC52	FastEthernet0	192.168.12.143	255.255.255.192	192.168.12.129
PC53	FastEthernet0	192.168.12.194	255.255.255.192	0.0.0.0
Laptop1	Wireless0	192.168.12.145	255.255.255.192	192.168.12.129
Laptop2	Wireless0	192.168.12.147	255.255.255.192	192.168.12.129
Laptop3	Wireless0	192.168.12.144	255.255.255.192	192.168.12.129
Printer1	FastEthernet0	192.168.11.142	255.255.255.192	192.168.11.129
Printer2	FastEthernet0	192.168.11.202	255.255.255.192	192.168.11.193
Printer3	FastEthernet0	192.168.12.11	255.255.255.192	192.168.12.1
Printer4	FastEthernet0	192.168.12.88	255.255.255.192	192.168.12.65
Printer5	FastEthernet0	192.168.12.149	255.255.255.192	192.168.12.129
Server1	FastEthernet0	192.168.12.197	255.255.255.192	192.168.12.193
Server2	FastEthernet0	192.168.12.196	255.255.255.192	192.168.12.193
Server3	FastEthernet0	192.168.12.198	255.255.255.192	192.168.12.193
IP Phone 1				

Step-by-Step Implementation

1. Step 1: Installed HWIC-2T modules on both routers for serial connectivity. Used serial DCE cables between Router 1 and Router 2.



B3R

Physical Config CLI Attributes

Physical Device View

Zoom In Original Size

MODULES

- HWIC-1GE-SFP
- HWIC-2T**
- HWIC-4ESW
- HWIC-8A
- WIC-Cover
- GLC-LH-SMD

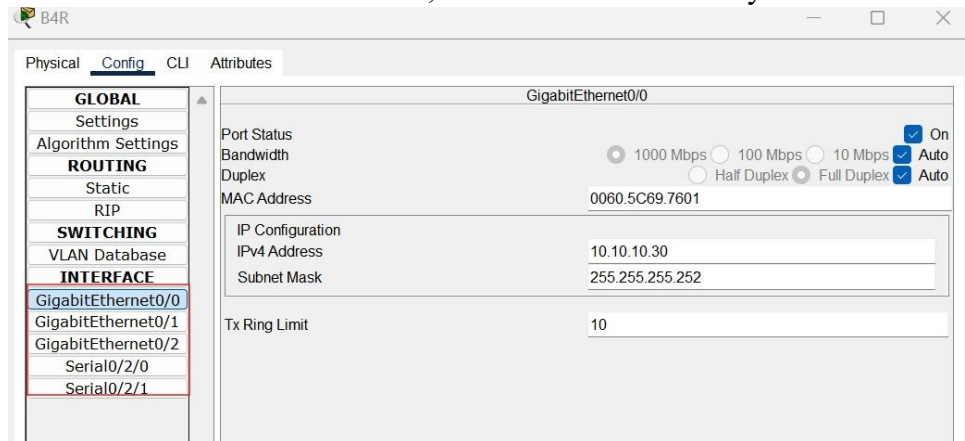
Network diagram showing connections between routers and switches:

- 2911 B3R (top left) connected to 3650-24PS L3 SW (top center) via 10.10.10.32/30.
- 2911 B3R connected to 3650-24PS B4L3 SW (bottom center) via 10.10.10.36/30.
- 3650-24PS L3 SW connected to 3650-24PS B4L3 SW via 10.10.10.40/30.
- 3650-24PS B4L3 SW connected to 2911 B4R (bottom left) via 10.10.10.24/30.
- 3650-24PS B4L3 SW connected to 2960-24P L3 SW (bottom right) via 10.10.10.44/30.
- 2911 B4R connected to 2960-24P L3 SW via 10.10.10.52/30.

Serial DTE



- Step 2: Configured static IP addresses on Router 1 and Router 2. Assigned IP addresses to the router interfaces, and verified connectivity.



- Step 3: Configured static routing between Router 1 and Router 2 to ensure inter-router communication.

```

Password:
B3R>ena
Password:
B3R#conf t
Enter configuration commands, one per line. End with CNTL/Z.
B3R(config)#
B3R(config)#int serial0/2/1
B3R(config-if)#ip add 10.10.10.37 255.255.252
                        ^
% Invalid input detected at '^' marker.

B3R(config-if)#ip add 10.10.10.37 255.255.255.252
B3R(config-if)#no shutdown
B3R(config-if)#
  
```

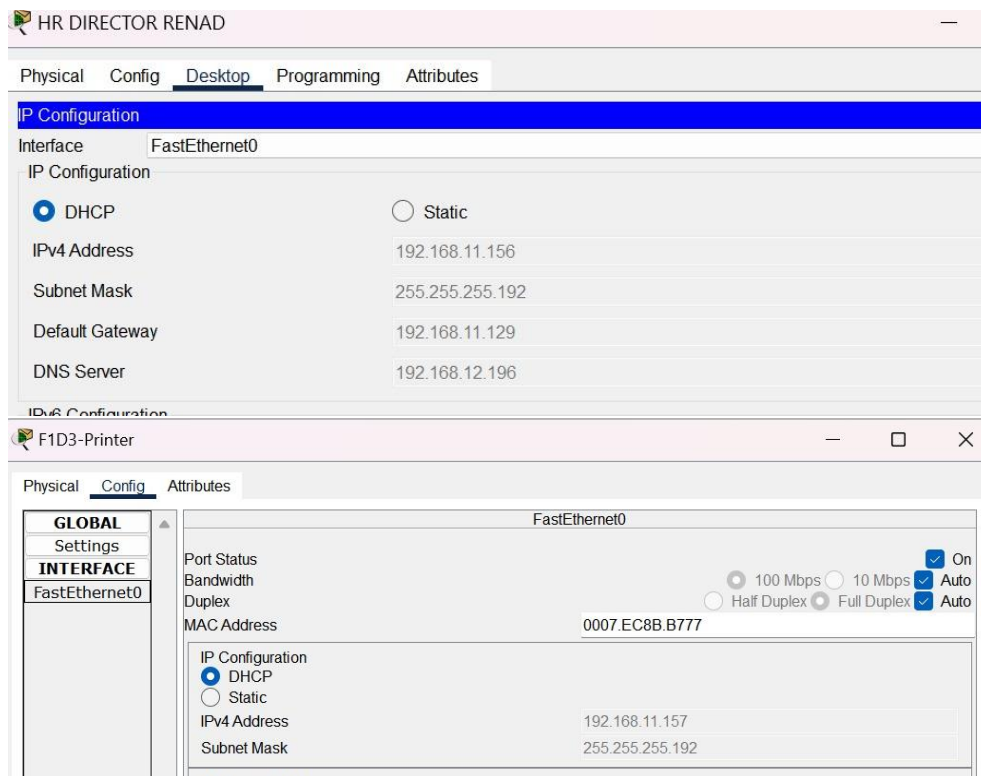
- Step 4: Set up DHCP on Router 1 to provide dynamic IP addressing to connected devices. Configured IP pools and DNS settings.

```

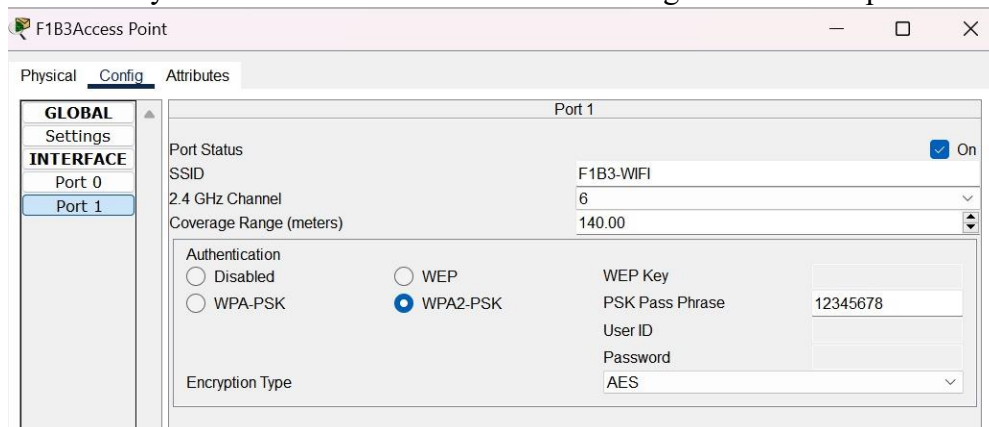
B3R(config-if)#no shutdown
B3R(config-if)#ip dhcp pool LAN_POOL
B3R(dhcp-config)#network 10.10.10.0 255.255.255.0
B3R(dhcp-config)#default-roter 10.10.10.37
                        ^
% Invalid input detected at '^' marker.

B3R(dhcp-config)#default-router 10.10.10.37
B3R(dhcp-config)#dns-server 8.8.8.8
B3R(dhcp-config)#exit
B3R(config)#ip dhcp excluded-address 10.10.10.37
B3R(config)#
  
```

- Step 5: Connected PCs and printers to the respective switches (Switch 1 for Router 1 and Switch 2 for Router 2). Assigned static IPs to devices where necessary.



6. Step 6: Configured an Access Point connected to Switch 1, providing secure wireless connectivity. Wireless devices were connected using the SSID and password.



7. Step 7: Connected IP Phones to the switches and configured them for VoIP. Phones were assigned IPs and verified for connectivity.
8. Step 8: Implemented VLANs on the switches: VLAN 10 for Data, VLAN 20 for Voice. Assigned appropriate ports to each VLAN and set trunking for VLAN communication.



```

FLOOR2B3SW
Physical Config CLI Attributes
IOS Command Line Interface

FLOOR2B3SW(config)#vlan 70
FLOOR2B3SW(config-vlan)#name hr
FLOOR2B3SW(config-vlan)#vnam 80
^
% Invalid input detected at '^' marker.
FLOOR2B3SW(config-vlan)#na,r sales
^
% Invalid input detected at '^' marker.
FLOOR2B3SW(config-vlan)#name sales
FLOOR2B3SW(config-vlan)#vlan 100
FLOOR2B3SW(config-vlan)#name managment
FLOOR2B3SW(config-vlan)#vlan 110
FLOOR2B3SW(config-vlan)#name finance
FLOOR2B3SW(config-vlan)#int fa0/2
FLOOR2B3SW(config-if)#switchport mode access
FLOOR2B3SW(config-if)%%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk
FastEthernet0/2 VLAN1.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/2 on VLAN0001. Inconsistent port type.

FLOOR2B3SW(config-if)#switchport access vlan 70
FLOOR2B3SW(config-if)#switchport mode trunk

FLOOR2B3SW(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

FLOOR2B3SW(config-if)#switchport trunk allowed vlan 70,80,90,100,110
FLOOR2B3SW(config-if)#
  
```

9. Step 9: Tested the network by pinging devices across subnets and verifying connectivity. DHCP was tested for dynamic IP assignment.

```

HR NORAH
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.11.205

Pinging 192.168.11.205 with 32 bytes of data:

Reply from 192.168.11.205: bytes=32 time=1ms TTL=127
Reply from 192.168.11.205: bytes=32 time<1ms TTL=127
Reply from 192.168.11.205: bytes=32 time=1ms TTL=127
Reply from 192.168.11.205: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.11.205:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 169.254.197.150

Pinging 169.254.197.150 with 32 bytes of data:

Reply from 192.168.11.129: Destination host unreachable.
Reply from 192.168.11.129: Destination host unreachable.
Reply from 192.168.11.129: Destination host unreachable.
Reply from 192.168.11.129: Destination host unreachable.

Ping statistics for 169.254.197.150:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
  
```

Conclusion



The project achieved complete network connectivity with effective IP addressing, routing, DHCP configuration, VLAN segmentation, and secure wireless access. All devices were successfully interconnected, and the network operated smoothly. This design demonstrates a comprehensive understanding of network design and configuration using Cisco Packet Tracer.